Math 534H - Spring 2026 - Additional Practice Problems.

1. Solve the transport equation $3u_x + 7u_y = 0$ with auxiliary condition $u(0,y) = 2y$

2. Solve the linear inhomogeneous equation

$$2u_x + 5u_y = 4$$

3. a) Use the coordinate method (as in Lecture 5) to find the general solution of $3u_x - u_y - 2u = 0$

b) Find the solution that also satisfies $u(x,0) = e^x$

4. Use the coordinate method to find the general solution of $u_x - u_y + 2u^2 = 0$

(Hint: use $\xi = x+y$, $\eta = x-y$)

5. a) Solve the inhomogeneous transport equation

$$u_t + 3u_x = x+t$$

Hint: Solve homog. problem and use Duhamel formula as in Lecture 6. Note here $F(x,t) = x+t$.

b) Find the solution that satisfies $u(x,0) = 4x$.

6 Find the solution $u(x,y)$ to the transport equation $\frac{1}{1+x^3} u_x + \frac{1}{1+y^2} u_y = 0$ such that $u(x,0) = x + \frac{1}{3}x^3$.

(<u>Hint</u>. Use separation of variables to solve the ODE giving the characteristic curves; specify the region of validity and work with the solution to the ODE $y = y(x)$ in intrinsic form. This procedure is what we did in Lecture 4; see also Strauss's section 1.2 pages 8-9).

7 Find the regions of the (x,y) plane where the given second order linear PDE below is elliptic, hyperbolic or parabolic. Justify your answer and find its characteristic curves:

$$3y u_{xx} - 6u_{xy} + 3x u_{yy} = 0.$$

8 a) Check $u_{xx} + 2u_{xt} - 80u_{tt}$ is hyperbolic.

b) Solve the following initial value problem (IVP)

$$(*)\quad \begin{cases} u_{xx} + 2u_{xt} - 80u_{tt} = 0 \\ u(x,0) = 4x \\ u_t(x,0) = 1 \end{cases}$$

by first completing the squares in the equation, do a change of variables and undo to find $u(x,t)$ as the sum of general differentiable functions f and g of one variable suitably evaluated. Then use the initial conditions to choose appropriate f and g and determine constants (as we did in Lecture 10 example 3). Check your solution satisfies $(*)$.

9 Consider the solution to the following IVP for the wave equation on $\mathbb{R} \times [0,\infty)$:

$$\begin{cases} u_{tt} - u_{xx} = 0 \\ u(x,0) = 0 \\ u_t(x,0) = \chi_{[-1,1]}(x) \end{cases}$$

Recall

$$\chi_{[-1,1]}(x) = \begin{cases} 1 & \text{if } |x| \le 1 \\ 0 & \text{otherwise} \end{cases}$$

a) Use D'Alambert's formula to write down the solution $u(x,t)$ in terms of its initial data. (do not evaluate the integral)

b) Use basic rules of differentiation (chain rule in t. Recall Theorem 3 in Appendix 3 of Strauss) to compute $u_t(x,t)$

c) Set $x=0$ in b) and prove that $u_t(0,t) = 0$ for all $|t| > 1$.

10 Let $u(x,t)$ be a smooth to the <u>damped</u> string equation:

(†) $u_{tt} - u_{xx} + 7u_t = 0 \quad x \in \mathbb{R},\ t > 0$

with initial conditions $u(x,0) = \phi(x)$ and $u_t(x,0) = \psi(x)$ where ϕ and ψ are smooth functions decaying fast to zero as $|x| \to \infty$ (this assumption guarantees that all integrals involved below are finite).

Let $E[u](t) := \frac{1}{2}\int_{\mathbb{R}} |u_t|^2 + |u_x|^2 \, dx$ be

Note this is $\geq 0 \ \forall t$.

these are functions of (x,t).

the energy associated to (†)

a) Assume also that $u_x(x,t) \longrightarrow 0$ as $|x| \to \infty$ and prove that $E[u](t) \leq E[u](0)$.

b) Use a) to prove the uniqueness of solutions to the IVP above.